BODY SUPPLY STATION

Modular Assembly System (MAS 200)

COMPUTER-INTEGRATED MANUFACTURING @ E120

1. TECHNICAL AND FUNCTIONAL DESCRIPTION

From SMC, a world leader in Pneumatics, we introduce **SMC International Training**. Our clear and wholehearted international vocation defines the objectives of this company: the conception and marketing of training systems in the automation field to satisfy the training needs of educative centers and companies all over the world.

An example of a Comprehensive Training System developed within the framework of this project, constitutes the Modular Assembly System MAS-200 (figure 1).

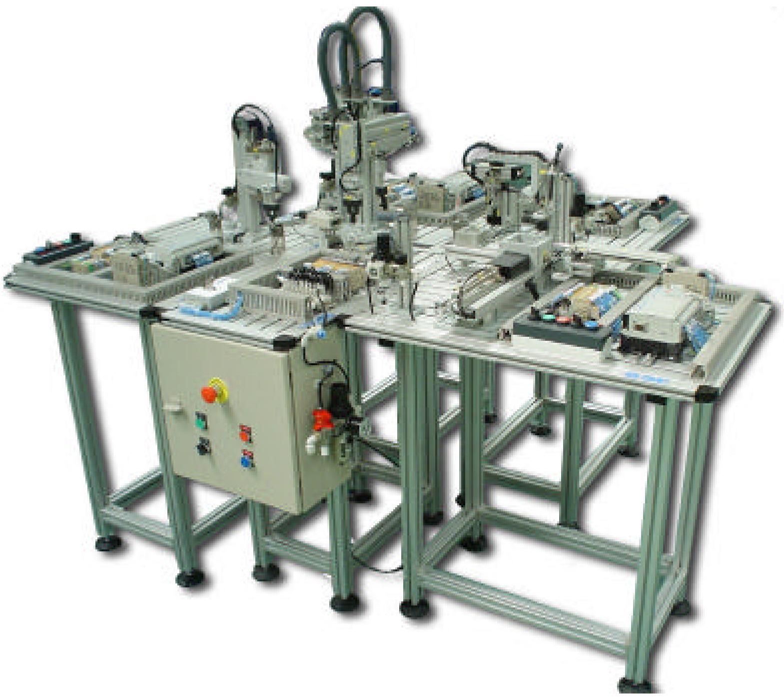


Figure 1: Modular Assembly System MAS-200

The modular assembly cell has been specially conceived for persons to acquire professional capabilities in connection with the Occupational Groupings of Electricity/ Electronics and Maintenance, such as:

- Installation, Electromechanical Maintenance and Line Transport.
- Industrial Equipment Maintenance.
- Automatic Control and Regulation Systems.

It enables the development of various skills associated with pneumatic, electro-pneumatic, electrical, robotic and handling automatisms, programming and PLC technologies, industrial communications, supervision, quality control and fault diagnosis and repair. It also allows the study of a wide range of sensor types:

- Magnetic detectors.
- Inductive detectors.
- Photoelectric detectors.
- Reed detectors.
- Vacuum switches.
- Bar Code readers.
- Linear encoders.
- Etc.

The system comprises a modular assembly cell which carries out an assembly process involving a number of predetermined parts (figure 3) that form a turning mechanism
(figure 2).

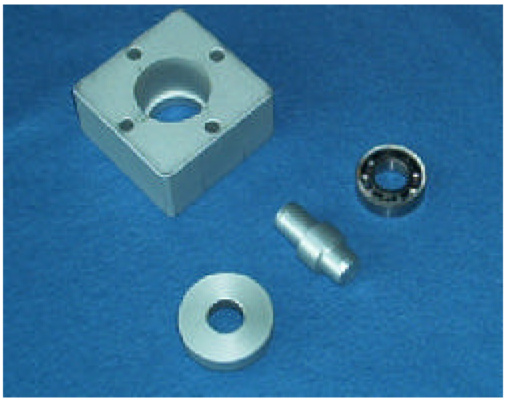


Figure 2: Turning mechanism

Figure 3: Turning mechanism components

Parts are transported between the different stations or layouts by an central station (components transfer station). Parts are mounted on the body in the final position of first station.

2. PROCESS STATIONS

The process stations or layouts function either independently of the transport system, or integrated into it. The stations are situated around the components transfer station, and may work in a completely independent and self-sufficient mode or integrated into the system, step by step or automatic mode.

Figure 4: Example of process station

Each station has its own electrical panel, where the wiring system and PLC are fully visible for study, while new elements may be fitted to the panel if desired. This electrical control panel may be made entirely at each station for use in programmable PLC training. In addition, students may design and build their own controls with different PLCs and subsequently integrate them in the station, thereby developing a further series of skills envisaged in the Training Cycles for those persons who form the target group for the System.

The top of each station incorporates the start, stop, reset and single and continuous cycle pushbuttons.

The system is modular and may be extended, allowing future incorporation of other process stations according to user needs.

The stations are mounted on aluminum sections, forming tables with a large surface area and multiple slots to allow all types of extension and modification.

The assembly process performed (turning mechanism) is as follows:

Layout 1: Feed body to which the other parts are assembled.
Layout 2: Bearing manipulation station (Pick and Place bearing).
Layout 3: Shaft insertion station (Pick and Place shaft).
Layout 4: Cover positioning station (Pick and Place cover).
Layout 5: Component transfer station, with two version: electropneumatic or by robot.

PLC WITH NETWORK COMMUNICATIONS SYSTEM.

The standard layout includes a **PLC** panel-mounted via a DIN track.

The **PLC** has inputs and outputs, timed and programmed interrupts, internal PID, high-speed counters, etc. It incorporates a port for programming, and a port for communication with other PLCs or PCs.

The **PLC**s which govern the cell stations communicate with each other via a network, so that each station comes ready with the necessary adapter and communications module.

PLC PROGRAMMING SOFTWARE (OPTIONAL)

This software enables programming of instructions and contact diagrams for all standard PLCs in any desired layout. It allows ON LINE contact diagram monitoring and modification. There are options to generate listings with comments, data reports and memory listings. It incorporates help advice in all functions. The software includes an operating manual and works under the Windows environment.

2.1. Body supply station

2.1.1. Station function

This first station feeds in the body which is the support for the turning mechanism, and moves it to the final position in this very station, where the central station (components transfer station) assemblies the rest of the components of the turning mechanism (figure 5).

Figure 5: Body supply station

This operation begins when the start pushbutton is pressed (either in the central station of components transfer if the station is working in integrated mode (in network) or in the very station if the station is working in independent mode). The confirmation of the body is in correct orientation is got by a pneumatic cylinder like a checker turning on an internal flag of the PLC, signal used to begin assembling the rest of components from the other stations.

2.1.2. Integral parts

Station 1, like the others, may be divided into a series of modules. Each sub-division has been made by considering it as a set of components which performs a specific operation within the process completed at the station. Starting out from this consideration, a description is given below of the ordered sequence of actions performed for assembly of the body, indicating the components involved in each operation.

- ***Body feed:***

The feeder which supplies the body is of the gravity feed type, in that the bodies are stored in a stack so that when the bottom one is removed, the next falls into place under the effect of its own weight and that of the ones above it.

The body is extracted by a pneumatic cylinder which pushes against a pusher shaped to match the profile of the body.

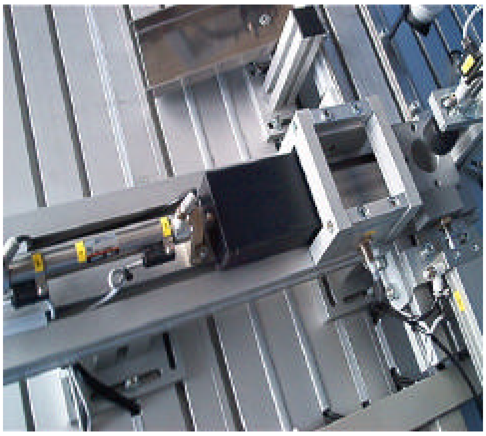


Figure 6: Body feeder

In this feeder, in the low place, there is an inductive sensor (figure 7). This inductive sensor is used to detect no material, in this way the PLC can verify when the bodies are finished and more bodies must to be put into the feeder.

Figure 7: Inductive sensor

- ***Position verification:***

The body contains a housing in which the other components are fitted. This housing must always be facing upwards when the base is placed on the final position of this very station. To check correct body orientation, a check is made by a cylinder which advances and inserts a cylindrical part in the body housing. If the body is inverted, this part cannot enter the housing, the cylinder cannot complete its stroke and the magnetic detector on the cylinder is not activated (figure 8).

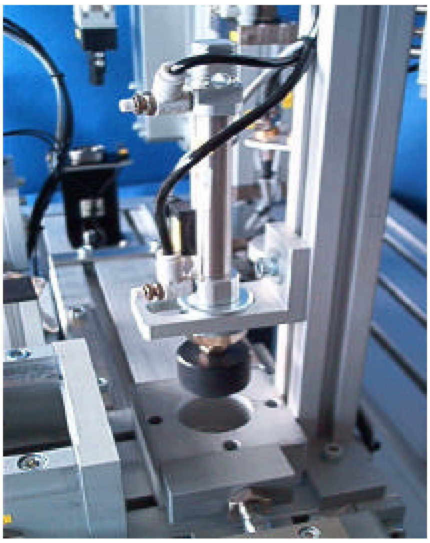


Figure 8: Position checker

A signal to this effect enters the PLC, resulting in an indication that the position of the body is incorrect.

- ***Movement to the final position:***

A cylinder with a pusher at its end is used to situate the body at the final position where the rest of the components are assembled into its inside.

The cylinder is rectangular, to prevent the pusher from turning.

- ***Body rejection:***

If the verification process shows the position of the body to be incorrect, a single-acting cylinder moves it towards a ramp, leaving the space clear for a new body.

When the assembly and disassembly process of the turning mechanism components is finished, the assembly position must be free too, of this way, a new turning mechanism can be assembled. In this case, the single-acting cylinder works too, it moves the body towards the ramp. See figure 9.

Figure 9: Rejection of body

2.1.3. Technical data

Dimensions:

- Table of slotted aluminum section, 840 x 580 mm. Height 870 mm.

Air treatment unit:

- Filter to 5 µm, pressure regulator and pressure gauge.

Pushbutton control:

- Start, stop, reset pushbuttons. Cutout selector, mode selector (auto-man) and error pilot indicator.

Composition of station modules:

- Body feed module
  - Magazine capacity: 12 bodies.
  - Actuators:
    - Double-acting pusher cylinder Ø16, Stroke: 100mm (CD85N16-100B), with flow regulators and initial-end position detector. Controlled by 5/2 way monostable solenoid valve.
  - Sensors:
    - Reed type magnetic detectors (D-C73L).
    - Inductive sensor (OMRON E2A-M12KS04WPB1).

- Position verification module
  - Actuators:
    - Double-acting cylinders Ø12, Stroke: 50mm (CD85N12-50A), with flow regulators and end position detector. Controlled by 5/2 way monostable solenoid valve.
  - Sensors:
    - Reed type magnetic detector (D-A73CL)

- Movement to the final position module
  - Actuators:
    - Rectangular section pusher cylinder Ø25, Stroke: 200mm (MDUB25-200DMZ), with flow regulators and end position detector. Controlled by 5/2 way monostable solenoid valve.
  - Sensors:
    - Reed type magnetic detector (D-A93L).

- Body rejection module
  - Actuators:
    - Single-acting ejection cylinder Ø10, Stroke: 15mm (CJPB10-15H6). Controlled by 3/2 way monostable solenoid valve.

Electrical control panel:

- Accessible terminal plate with supply connections and coded I/Os.
- I/O station: 9 inputs, 5 outputs.
- Supply module: 24V/60W
- PLC control:
  - CPU
  - Communication module for PLC network connection.

3.1.3. Air requirements

Details are given below of the air requirements for each station and for the whole cell:

Station 1: Body supply station:

Element	Cycles	Diameter	Stroke	Consum l/cycle
CD85N16-100B	2	16	100	0,16
CD85N12-50B	2	12	50	0,0144
MDUB25-200DMZ	2	25	200	0,7852
CJPB10-15H6	1	10	15	0,0044
CYCLE TIME	20 sec			
STATION 1 TOTAL	**3 l/min**			

Station 2: Bearing manipulation station:

Element	Cycles	Diameter	Stroke	Consum l/cycle
MSQB50A	2	-	-	0,32
MHK2-16D	2	16	10	0,028
CYCLE TIME	10 sec			
STATION 2 TOTAL	**2,1 l/min**			

Station 3: Shaft insertion station

Element	Cycles	Diameter	Stroke	Consum l/cycle
MHK2-16D	2	16	10	0,028
EMRQBS32-25CB	4	32	25	0,3214
	2	-	-	0,084
CYCLE TIME	10 sec			
STATION 3 TOTAL	**2,6 l/min**			

Station 4: Cover positioning station

Element	Cycles	Diameter	Stroke	Consum l/cycle
CXSM15-100	2	15	100	0,1413
CXSM10-50	4	10	50	0,0628
ZU07S	19 l/min			
CYCLE TIME	10 sec			
STATION 4 TOTAL	**20 l/min**			

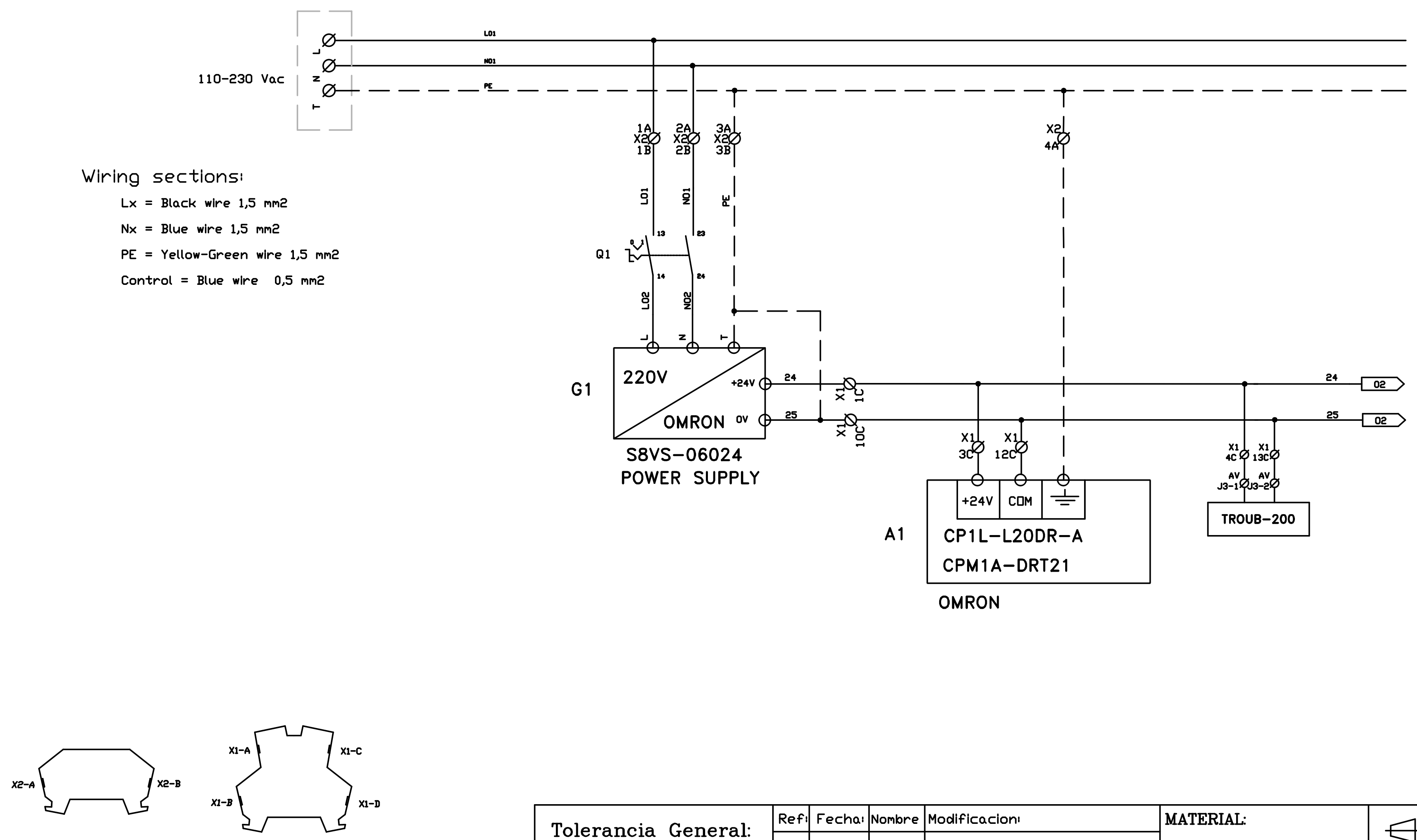
Station 5 (version 1): Component transfer station

Element	Cycles	Diameter	Stroke	Consum l/cycle
CDQ2B25-40D	16	25	40	1,256
CQ2B16-10D	32	16	10	0,2572
MGPM12-20	2	12	20	0,0058
MGPM20-150A	12	20	150	2,2608
CXSM15-50	12	15	50	0,4236
MHK2-16D	2	16	10	0,028
MHKL2-16D	4	16	10	0,056
CYCLE TIME	150 sec			
STATION 5 TOTAL	**1,7 l/min**			

MAS-200
USER MANUAL

ANNEX B: ELECTRIC DIAGRAMS

INTERNATIONAL TRAINING



Wiring sections:

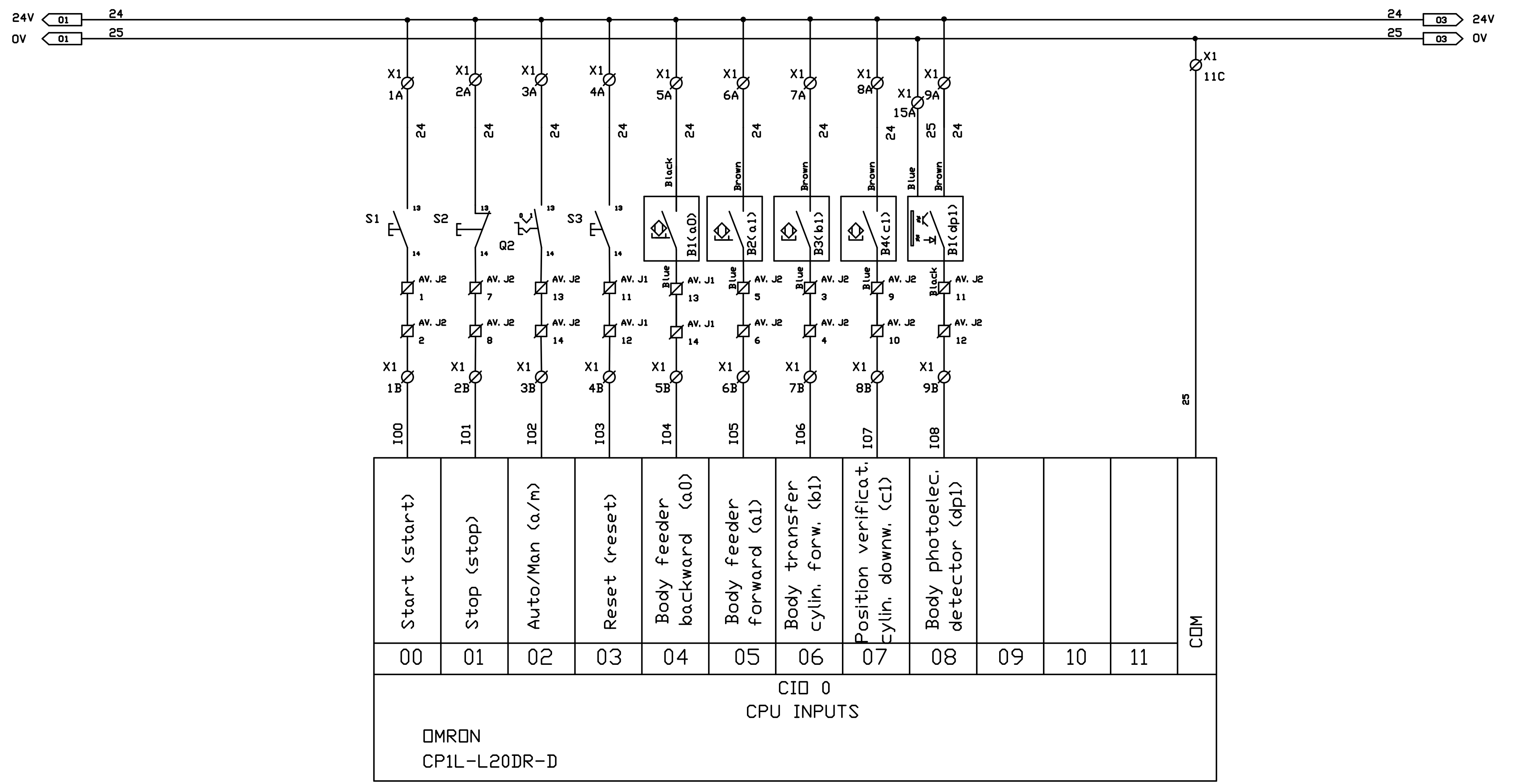
Lx = Black wire 1,5 mm2

Nx = Blue wire 1,5 mm2

PE = Yellow-Green wire 1,5 mm2

Control = Blue wire 0,5 mm2

Tolerancia General: Segun ISO-2768 m	Ref:	Fecha:	Nombre	Modificacion:	MATERIAL:		ESCALA: Free
SMC INTERNATIONAL TRAINING					TRATAMIENTO	Fecha: 09-07-2014	
					CANTIDAD:	Dibujado: Aitor	
						Comprobado:	
	MAS-201				DESIGNACION: POWER CIRCUIT BODY SUPPLY MODULE	Nº DE PLANO: A/1-1	



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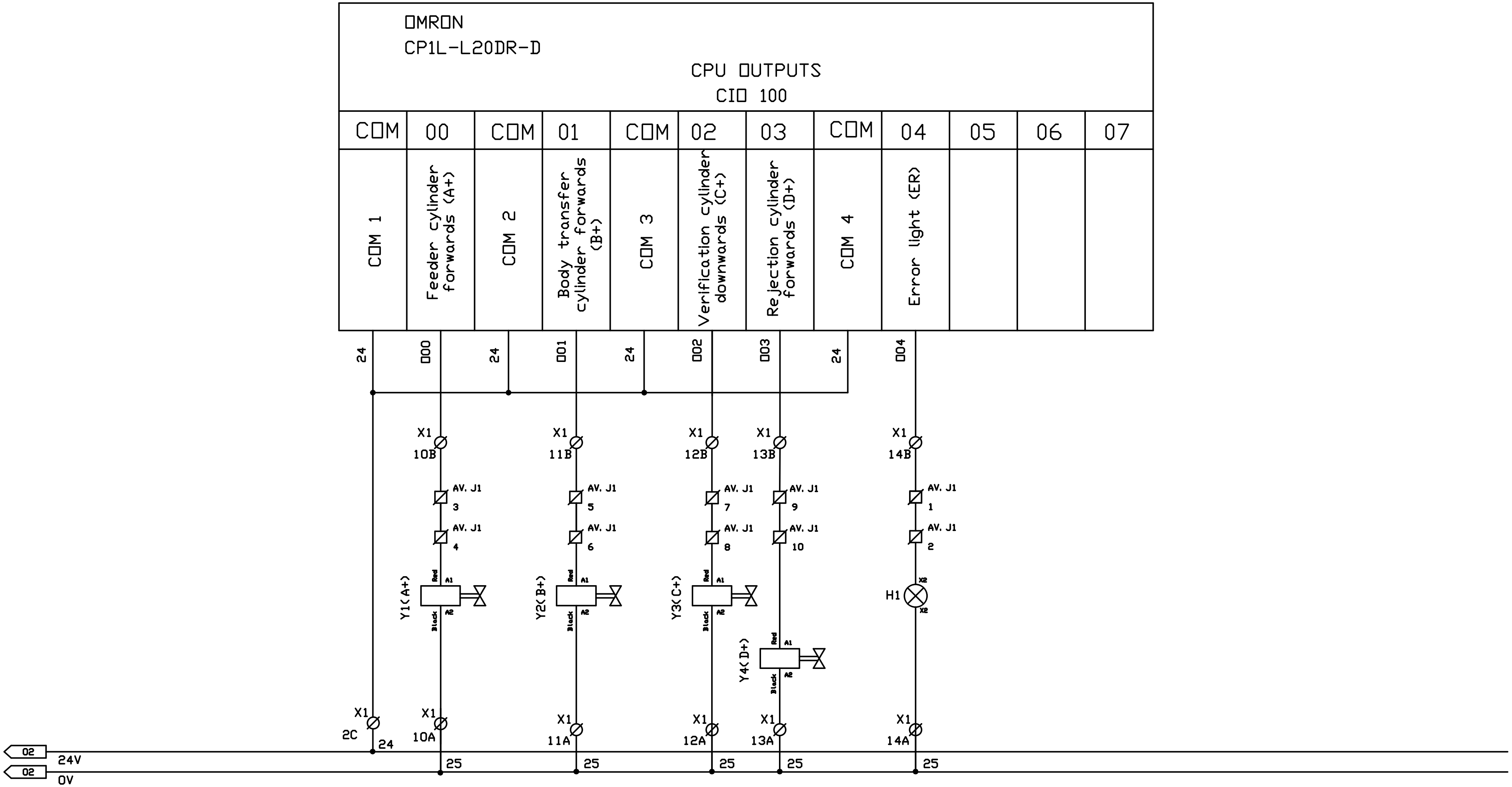

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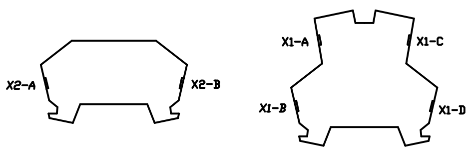
					CANTIDAD:	Dibujado: Aitor	
						Comprobado:	
SMC INTERNATIONAL TRAINING	MAS-201				DESIGNACION: CONTROL CIRCUIT OUTPUTS BODY SUPPLY MODULE	Nº DE PLANO: A/1-3	

X1

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
B	I00	I01	I02	I03	I04	I05	I06	I07	I08	□00	□01	□02	□03	□04	
A	24	24	24	24	24	24	24	24	24	25	25	25	25	25	25
C	24	24	24	24						25	25	25	25		
D															

X2

1	2	3	4
L01	N01	PE	PE
L01	N01	PE	

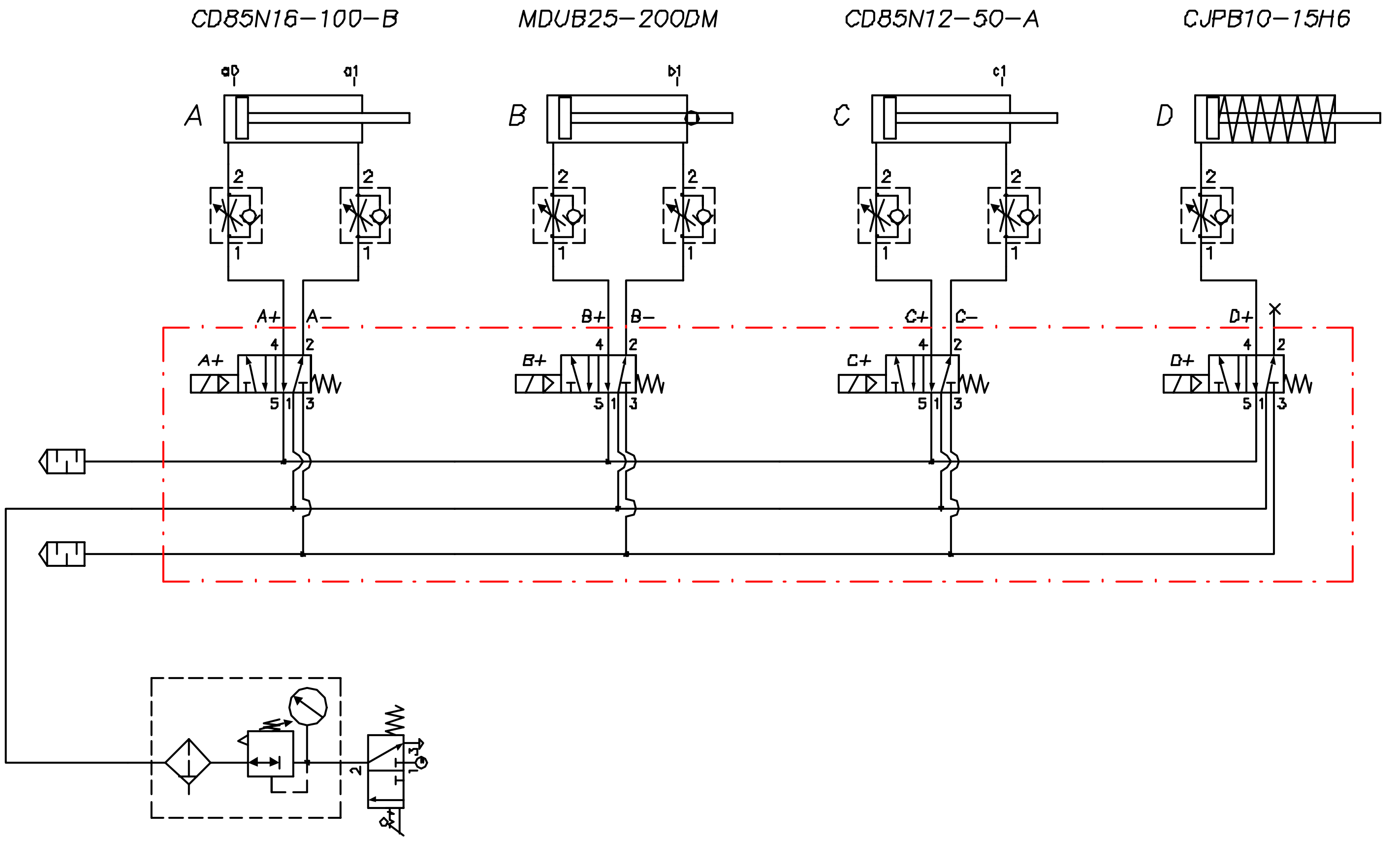


Tolerancia General: Segun ISO-2768 m	Ref	Fecha	Nombre	Modificacion	MATERIAL:	ESCALA: Free
SMC INTERNATIONAL TRAINING					TRATAMIENTO	Fecha: 09-07-2014
					CANTIDAD:	Dibujado: Aitor
						Comprobado:
	MAS-201				DESIGNACION: TERMINALS BODY SUPPLY MODULE	Nº DE PLANO: A/1-4

MAS-200
USER MANUAL

ANNEX C:
PNEUMATIC
DIAGRAMS

INTERNATIONAL TRAINING



MAS 200

BODY SUPPLY STATION

PNEUMATIC DRAWING

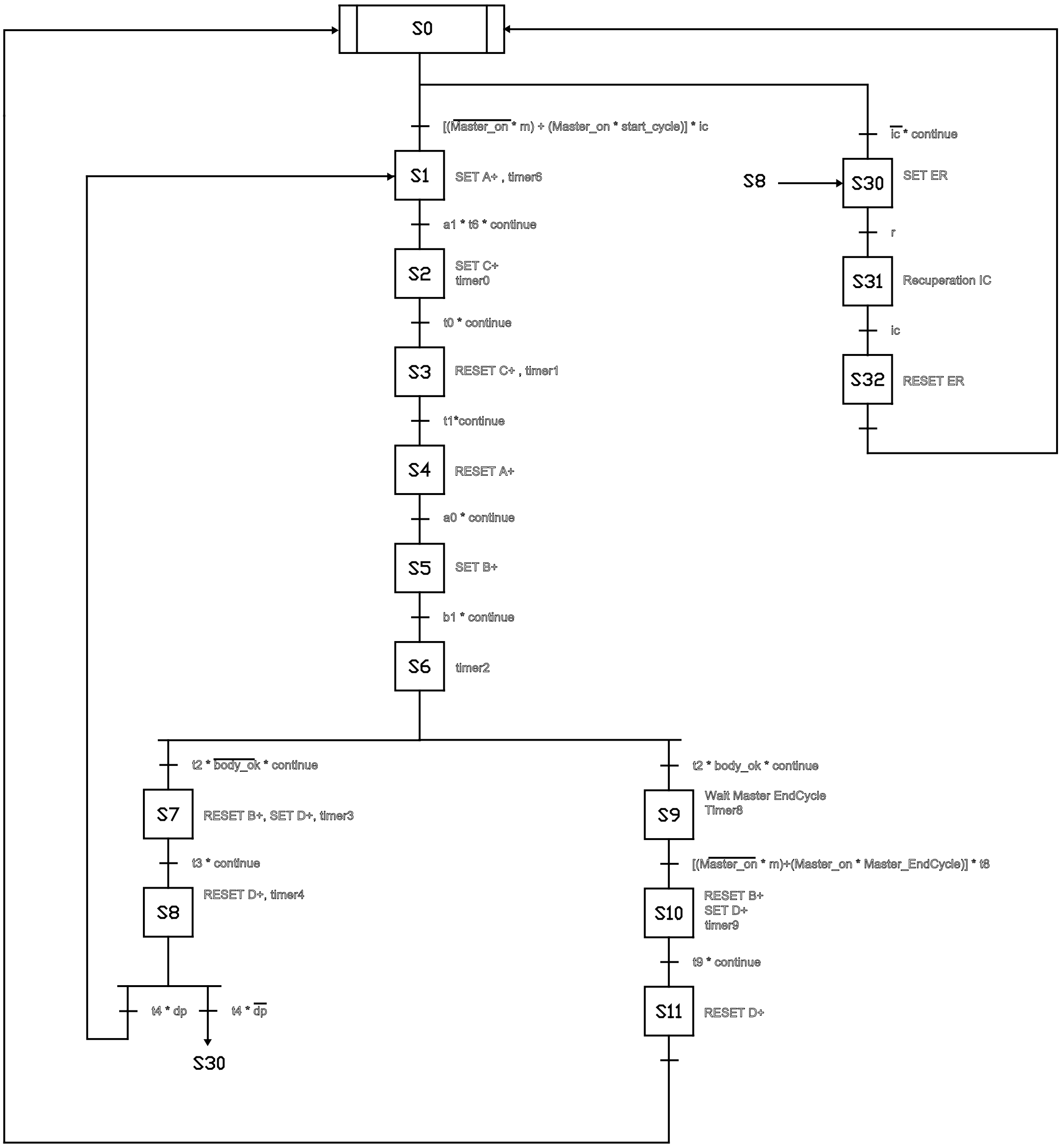
01

SMC INTERNATIONAL TRAINING

MAS-200
USER MANUAL

ANNEX D:
GRAFCETS OF THE PROGRAMS

INTERNATIONAL TRAINING



Tolerancia General: Segun ISO-2768 m	Ref	Fecha	Nombre	Modificacion	MATERIAL:	ESCALA:
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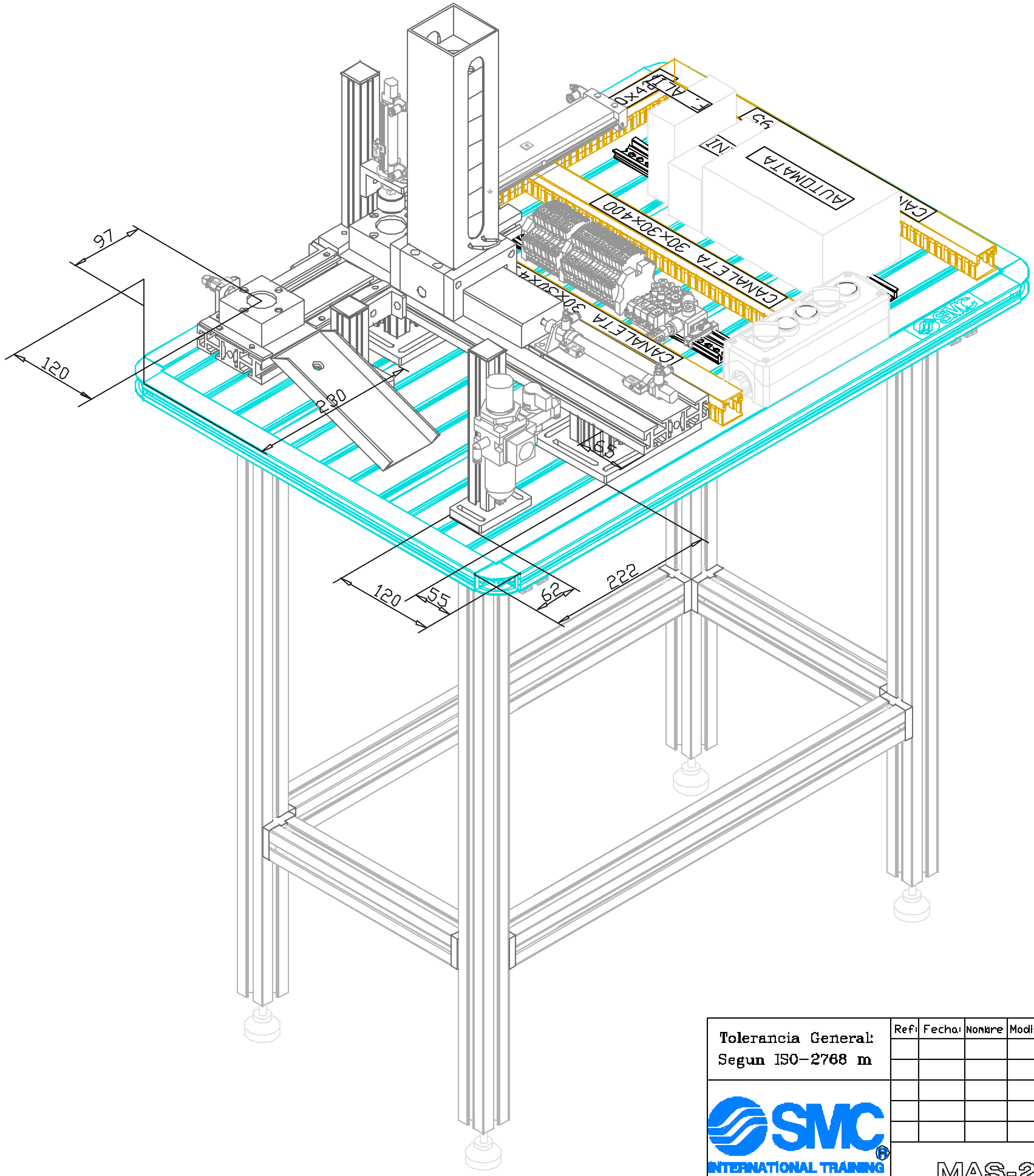
					CANTIDAD:	Dibujado: J.R.BARREAL Comprobado:
SMC INTERNATIONAL TRAINING	MAS-200			DESIGNACION: GRAFCET MAS200-BODY		Nº DE PLANO: 1-1

MAS-200
USER MANUAL

ANNEX E: MECHANICAL DIAGRAMS

INTERNATIONAL TRAINING



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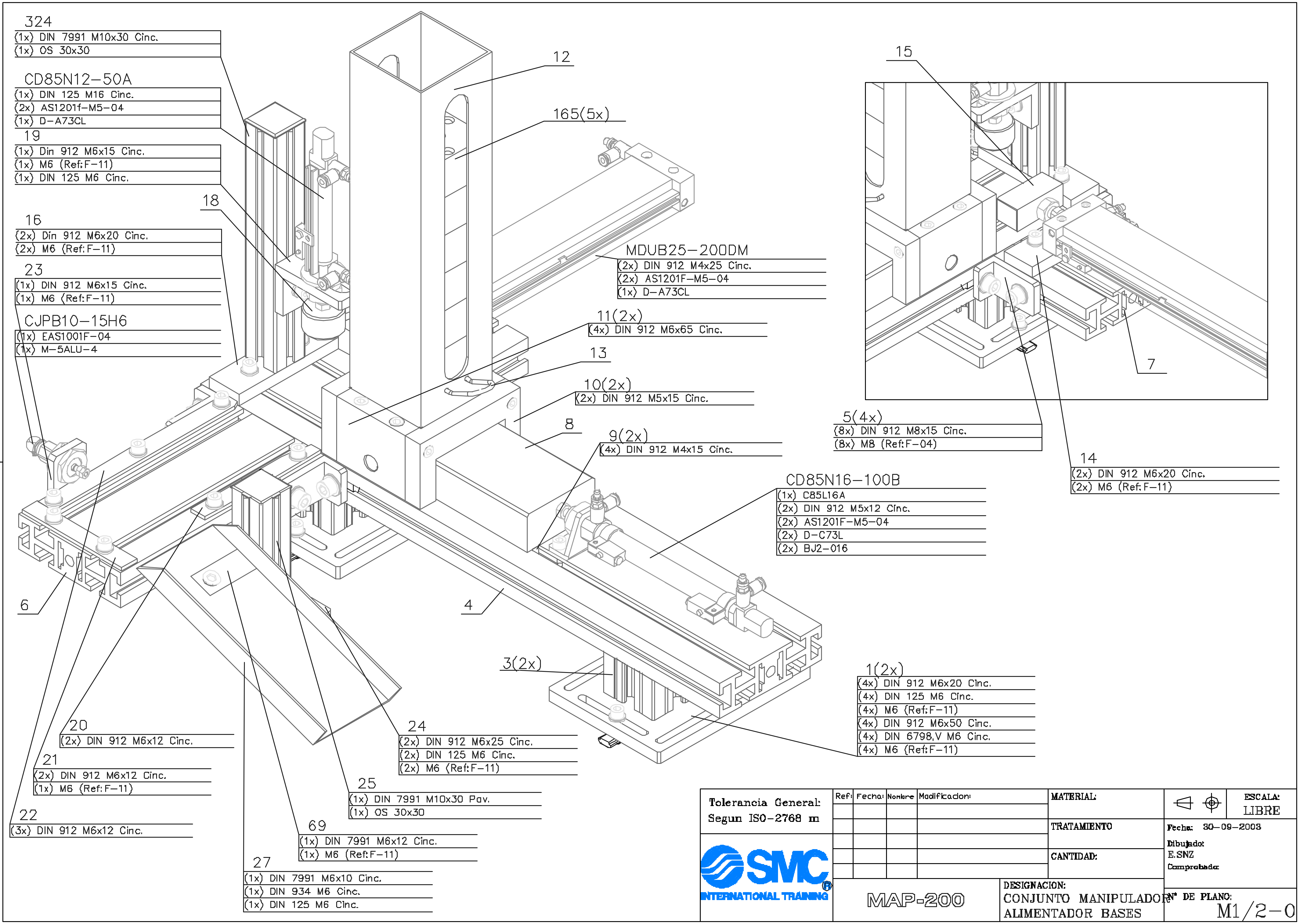
SMC INTERNATIONAL TRAINING					CANTIDAD:	Dibujado: E.SNZ Comprobado:
	MAS-200				DESIGNACION: CONJUNTO MAS-201	Nº DE PLANO: MS1/0-0



Tolerancia General:
Segun ISO-2768 m

SMC INTERNATIONAL TRAINING

Ref:	Fecha:	Nombre	Modificacion:

MAP-200

MATERIAL:

TRATAMIENTO

CANTIDAD:

DESIGNACION:
CONJUNTO MANIPULADOR
ALIMENTADOR BASES

ESCALA:
LIBRE

Fecha: 30-09-2003

Dibujado:
E.SNZ

Comprobado:

Nº DE PLANO:
M1/2-0